A Metric Details

This appendix expands the metrics defined in §3.5.

Accuracy grading. Accuracy is the primary ranking metric, mirroring InteractComp, BrowseComp, and FinanceBench. A GPT-4o-mini grader marks a final answer correct against the intended interpretation under financial-domain tolerance: a $\pm 1\%$ numeric tolerance, entity/ticker equivalence, and currency normalization, with a fiscal-year mismatch counted as wrong.

Interaction targeting (AC@1). For each interact action, a GPT-4o-mini classifier decides whether the clarifying question targets a true ambiguity category for that instance. We report the targeting accuracy AC@1 alongside the any-category, wrong-category, and generic-ask rates. Because a single clarification is often compound (one question can name an entity, a period, and a metric at once), AC@1 has two forms: a lenient *touch* form (the question names any true category) and a strict *central* form (the single most-central category is a true category). The touch form is validated against human annotators (§3.4). These diagnostics characterize *how* a model interacts and do not rank models.

Calibration. The Expected Calibration Error (ECE) between stated confidence and correctness uses five equal-width confidence bins.

Clarification efficiency (DisE^+). As a single supplementary number folding accuracy and clarification cost together, we report the efficiency of *successful* clarification,

$$\text{DisE}^+ = \mathbf{1}[\text{correct}] \times \mathbf{1}[n_{\text{asks}}>0] \times \frac{H_0}{n_{\text{asks}}},$$

which is positive only when the agent asked at least one question and then answered correctly, and decreases with redundant asks. We restrict it to the asked-and-correct case to avoid the perverse incentive of an all-instances variant, which would award its maximum to a confident *zero-ask* answer, rewarding the very behavior the benchmark penalizes. DisE^+ is a diagnostic of clarification efficiency, never the ranking metric.

B Detailed Results for the Core Findings

This appendix gives the full tables, figures, and findings summarized in §4.3, §4.2, and §4.4.

B.1 The Single-Gold Illusion

Before analyzing model behavior, we establish the core motivation empirically: *a conventional single-gold financial-QA benchmark cannot see the failure FinInteract measures.* Existing benchmarks (FinanceBench, FinQA, DocFinQA) attach one gold answer per question. Built from the same filings, such a benchmark would almost always adopt the *default* reading, the headline figure a non-expert assumes, as that gold. Our paired construction lets us quantify the consequence directly: every instance carries both an intended and a default answer, so we re-grade the *identical* model outputs under both conventions (Table 6(b)).

Finding 5: A single-gold convention over-states competence by up to $3\times$, and the gap is the benchmark's reason to exist. In the retrieval-augmented regime where current single-gold benchmarks operate, the larger models return the *default* reading more often than the intended one: GPT-4o is graded 37.0% correct against the default but only 12.1% against the intended answer, a single-gold benchmark would report $3.1\times$ its true competence. The model is not reasoning to the right answer; it is confidently producing a well-grounded answer to the *wrong reading of the question*, which a single-gold benchmark rewards and therefore cannot detect. (GPT-5-mini is the instructive exception: it captures *neither* reading because it asks rather than commits, so single-gold grading does not flatter it, but nor does it answer.) Critically, interaction is the *only* mode where intended accuracy exceeds default-capture (GPT-5: 20.2 vs. 12.1): asking is what converts a default-anchored guess into the correct intended answer. This is precisely the capability single-gold benchmarks are structurally blind to, and what FinInteract is built to measure.

Human validation of the default. The single-gold argument assumes a conventional benchmark builder, seeing only the question and passage, would record the *default* reading as gold. We test this directly: five annotators, shown Q and the passage but not the resolving context C, independently record the answer they would treat as gold. They converge strongly (93% of items unanimous; mean inter-annotator agreement 0.98) and select the *default* reading 66% of the time versus the intended reading 34%, with fewer than 8% reporting that they noticed a second defensible answer. A single-gold benchmark built from these filings would therefore encode the default, empirically confirming the illusion's premise rather than assuming it.

B.2 Latent Capacity and the Oracle Ceiling

To locate the bottleneck we bracket interactive accuracy with two ablations (Table 5 and Figure 4). **Forced interaction** requires the agent to ask $n{=}4$ clarifying questions before answering. **Context-oracle** hands the model the disambiguating context C (which fixes the intended interpretation) together with a passage containing *both* the intended and the default evidence spans, then asks it to answer, the ambiguity is pre-resolved but the model must still ground the correct value.

Finding 6: Models can answer once the ambiguity is resolved, they cannot resolve it themselves. The context-oracle ceiling is 93–95% for all three models, including GPT-5-mini, which scores 0% under free interaction. Latent answering capacity is therefore near-ceiling and roughly scale-invariant; what varies, and what FinInteract measures, is whether a model can *resolve* the ambiguity end to end, eliciting the disambiguating context and then integrating it into a committed answer. Crucially, *forcing* clarification does not recover this capacity: requiring four asks leaves GPT-4o and GPT-5-mini near 0% (0.0% and 0.6%) and *lowers* GPT-5 from 20.2% to 1.5%, because forced asks consume the round budget on low-value questions without converging on a committed answer. This already points past pure elicitation: models frequently ask on-category yet still fail to commit the

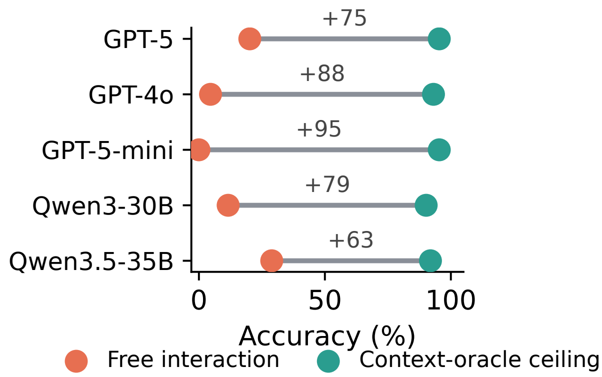

Figure 4: **The elicitation gap.** Free-interaction accuracy (orange) versus the context-oracle ceiling (teal) per model. Once the ambiguity is resolved every model answers 90–95% correctly, so the 63–95 point gap is the cost of the model failing to *elicit* the disambiguation itself, not missing knowledge.

right value (the dominant error mode, §C.7), so the bottleneck is targeted, self-initiated disambiguation *and* the integration of the resulting answer, which neither scale nor brute-force asking supplies.

B.3 Per-Category Skills

The five categories are not just data strata; each names a distinct *skill*, the ability to recognize that a question is underspecified *along that category* and ask the question that resolves it. We make this measurable by decomposing each model's per-category `+interact` competence into three components: **Recognition** (interaction rate, does it ask when it should?), **Targeting** (AC@1, does it ask about the *right* category?), and **Resolution** (accuracy, does asking yield the intended answer?). A model "has" a category skill only when all three are high; a low value localizes the failure to not-asking, asking-wrong, or asking-right-but-still-wrong. Table 6(c) reports Targeting, the decisive component, and Figure 5 shows the same profile as a per-category bar chart.

Finding 7: Ambiguity clarification is a per-category skill with a sharp touch versus central-targeting dissociation. Under the corrected classifier the pattern is precise. `recognition_policy` is a robust blind spot. No model ever asks about it, in either the touch or the central form (AC@1 $= 0$), so models essentially never question whether a figure is recognized point-in-time or over time. `temporal_scope` is fully solved. Models both touch it and make it the central concern (central AC@1 $= 1.0$). Entity and metric occupy a revealing middle. Models almost always *touch* them (touch AC@1 0.85–1.0) yet rarely make them the *central* point of the question (central AC@1 ≤ 0.16), defaulting instead to period confirmation. For these two categories the failure is not neglect but imprecision, the model names the entity or metric in passing while centering the question elsewhere. Resolution follows the same shape across all five models (GPT-5 resolves 57% of temporal but $\leq$ 23% of entity and metric, Figure 6). The gap is not uniform incompetence but missing specific targeting, which is precisely the intervention target (§C.7).

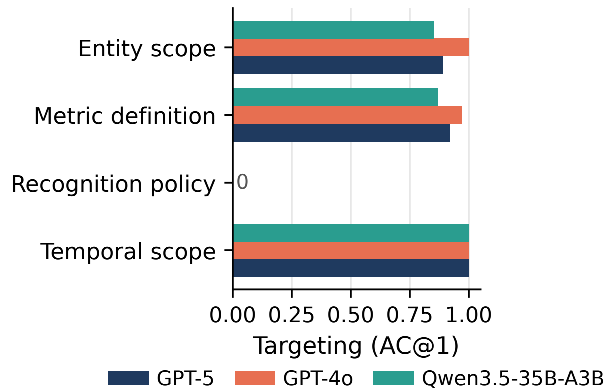

Figure 5: **Per-category clarification-capability profile** (touch AC@1 from Table 6(c)). Every model targets entity, metric, and temporal ambiguity well (0.85–1.00) but collapses to zero on `recognition_policy`, a blind spot shared across the closed frontier and the strongest open model.

Per-model, per-category case analysis. Placing targeting and resolution side by side for every model and category (Figure 6) surfaces three patterns. First, resolution does *not* track targeting: GPT-4o and GPT-5-mini touch the correct category on entity and metric almost as often as any model (AC@1 0.94–1.0) yet resolve those categories at 0–5%, so a high AC@1 is no guarantee of a correct answer. Second, the model ranking *reorders* across categories. Among the full-scale models Qwen3.5-35B-A3B leads on the two abundant categories (entity 40%, metric 32%) while GPT-5 leads on temporal (57%). Third, the seven proprietary models resolve entity and metric substantially better than the older OpenAI ladder (entity 21–48%, metric 26–42%, versus GPT-5's 23 and 13%), yet they inherit the *same* recognition blind spot, with touch AC@1 0 on every recognition instance they saw. `filing_vintage` is untestable everywhere (n=0). A single aggregate interaction score therefore conceals which category a given model can actually resolve, which is why we report per category.

C Additional Findings and Analyses

The main text reports four headline findings: the elicitation–integration gap, the single-gold illusion, the uneven per-category targeting, and the actionability of the ambiguity category as a training and inference signal. This appendix collects the remaining empirical findings and the full detail of the category-conditioned interventions.

C.1 The Illusion Is Not Finance-Specific: A General-Domain Replication

Finding 8: the single-gold illusion is domain-general. Running the identical default-vs-intended measurement on AmbigQA (Min et al. 2020) (Table 8) reproduces all three effects

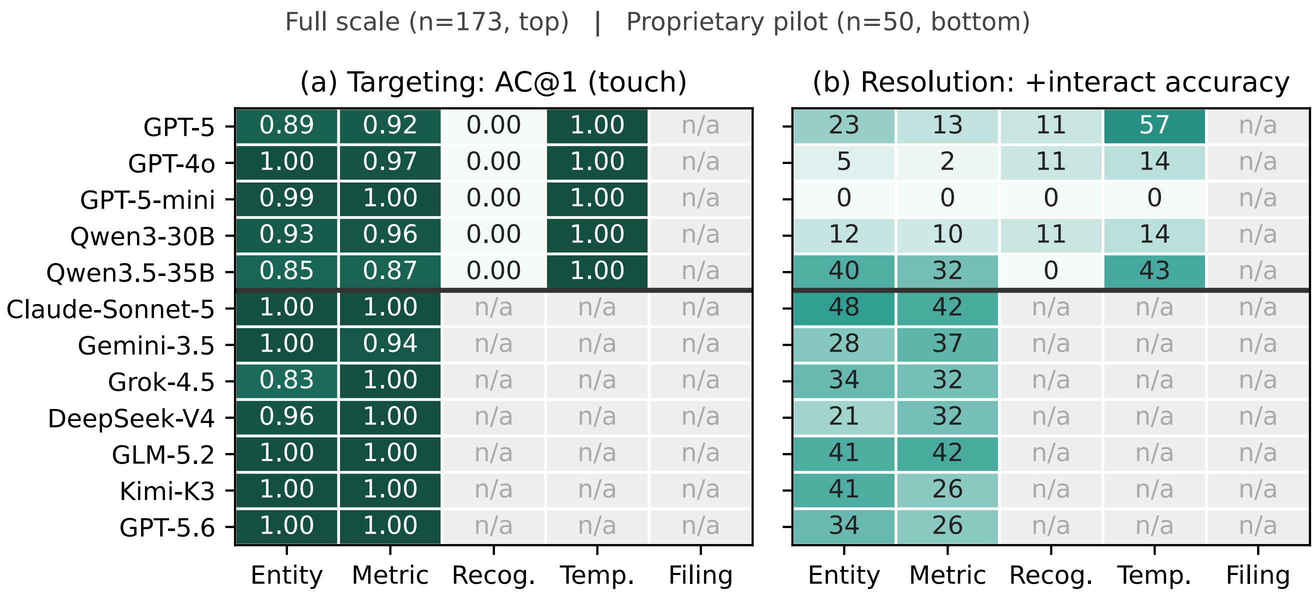

Figure 6: **Per-model, per-category performance.** *(a)* Targeting (AC@1 touch) versus *(b)* Resolution (`+interact` accuracy) for every model and ambiguity category. The top five rows are the full-scale panel (n=173, all categories) and the bottom seven are the proprietary pilot (n=50), for which only entity and metric carry enough instances (n=29 and 19), with recognition, temporal, and filing at $n \leq 1$ and shown n/a. `filing_vintage` is empty at full scale too (n=0). Every model targets entity and metric almost perfectly, the proprietary models resolve them better than the older OpenAI ladder, and no model targets recognition policy.

outside finance, *consistently across three models*. Grading the *same* answer-only outputs against any accepted reading rather than a specified target overstates competence by 2.1–2.3× (gpt-4o-mini 2.3, gpt-4o 2.2, gpt-5-mini 2.1), and the effect is robust to how the target reading is chosen, a random-target assignment gives 2.1×, versus our most-specific-reading default. Models return a valid but non-target reading 55–66% of the time, direct evidence the error is ambiguity-blindness, not noise. Elicitation is again underused: even permitted to ask, models clarify on at most 27% of ambiguous queries and usually far less (gpt-4o 2.5%, gpt-5-mini <1%), and interaction leaves intended accuracy far below the any-valid ceiling (gpt-4o-mini 43.5 vs. 81). The one honest difference is the oracle ceiling, 50–69% here versus 93–95% in finance, because open-domain factoids stay hard even once disambiguated, so general knowledge is itself a limiter. This is exactly why finance is the sharper instrument: its exact, injectable ground truth isolates *elicitation* from *knowledge* in a way general-domain QA cannot. The illusion itself, however, is a property of single-gold grading, not of finance.

C.2 Findings

We report results for the full OpenAI tier ladder, GPT-5, GPT-4o, and GPT-5-mini, across all three modes (Table 6). To probe whether the effect is an OpenAI-specific artifact, we additionally evaluate a preliminary seven-model, six-vendor panel of current frontier models (Claude-Sonnet-5, Gemini-3.5-flash, Grok-4.5, DeepSeek-V4, GLM-5.2, Kimi-K3, GPT-5.6) through a unified OpenRouter harness with an identical simulator/grader pipeline (N=50/mode, Table 6, panel (b)); it reproduces the headline pattern: added interaction is flat-or-negative for six of seven vendors and yields a clear gain only for Gemini, the weakest autonomous searcher. All headline accuracy comparisons below are reported with bootstrap 95% CIs over instances, and we test each within-model mode transition with a two-sided paired bootstrap (n=173). Six findings emerge here (Findings 9–14); the closely related latent-capacity ceiling is reported as a headline result in the main text (§4.2).

Finding 9: Interaction is a capability, not a free lever, it helps the strong model and *harms* the weak one. Adding interaction raises GPT-5 from 6.4% to 20.2% (95% CI [14.5, 26.6]; paired bootstrap p<0.001), but *lowers* GPT-4o from 12.1% to 4.6% (p=0.002). Forcing the weaker model to interact makes it significantly worse than not interacting at all. This extends InteractComp's "interaction stagnation" into a sharper claim: for models that cannot use it, interaction is actively harmful. Across all twelve evaluated models the interaction lever (added-interaction minus plain-search accuracy) is positive for only GPT-5 and Gemini and flat-or-negative for the other ten (Figure 7).

Finding 10: Model rankings flip with the interaction category. GPT-4o *outperforms* GPT-5 under pure retrieval (12.1 vs. 6.4) yet is *dominated* by it under interaction (4.6 vs. 20.2). A single-mode benchmark would mis-rank these models; the interaction category is what separates them, which is precisely the dimension FinInteract is designed to measure.

Table 8: The single-gold illusion replicates on AmbigQA (general domain, n=200, three models, *same* grading methodology; the grader retries with backoff so no answer is silently dropped). *Intended*: accuracy against a specified target reading. *Any-valid*: accuracy against *any* accepted reading, what a lenient single-gold benchmark credits. *Def.-cap*: fraction producing a valid but non-target reading. *Ask*: fraction that asks when permitted. gpt-5-mini interact omitted (it asks <1%). Modes: *A* answer-only, *A+O* answer with context oracle, *A+I* answer with interaction.

Model	Mode	Intended	Any-valid	Def.-cap	Ask
gpt 4o-mini	A	31.5	72.5	55.0	0
	A+O	49.5	68.5	31.0	0
	A+I	43.5	81.0	53.0	27.0
gpt 4o	A	39.0	84.0	65.5	0
	A+O	68.5	81.5	31.5	0
	A+I	43.0	86.0	64.0	2.5
gpt 5-mini	A	36.0	77.0	62.5	0
	A+O	62.0	73.5	27.5	0

Finding 11: Retrieval cannot substitute for interaction. Oracle retrieval lifts GPT-5 to only 6.4%: because each disambiguating passage contains *both* the intended and default values, retrieval surfaces the ambiguity but does not resolve it. Only a targeted clarification collapses the question to a unique answer, which interaction (20.2%) begins to do.

Finding 12: The weak model's failure is clarification non-integration, not inability to ask. GPT-4o asks readily (IR 99%, 4.7 questions on average) but fails to *integrate* the user's responses: 87% of its interactive errors are non-committal refusals (e.g. "please specify the exact period") issued *after* the simulator has answered. It can recognize ambiguity and ask, but cannot close the loop and commit, a multi-turn integration gap, not a protocol artifact (only 2% produced no answer at all).

Finding 13: Models are severely overconfident on ambiguous queries. Expected Calibration Error ranges from 0.53 (GPT-5, interact) to 0.95 (GPT-4o, answer-only); models report 85–95% confidence while answering 0–20% correctly. The remaining failure is not one of category *detection*. Under a human-validated classifier (agreement 0.75, matching inter-annotator agreement 0.73), both models almost always ask a question that touches the correct ambiguity category (AC@1 0.86 for GPT-5 and 0.94 for GPT-4o). What they lack is precise *central* targeting, only 0.06 and 0.04 of first questions make the true category their single central concern, and, above all, integration (Finding 12). The bottleneck is targeting precision and commitment rather tha category detection, which motivates the category-conditioned policy of §C.7.

Finding 14: A smaller model collapses into ill-formed clarification, it asks the most but commits the least. GPT-5-mini scores 0.6/0.0/0.0% across the three modes. The cause is not ambiguity-blindness but the opposite: it is the *most* eager to clarify (interact IR 96%, 4.6 asks/instance), yet it issues *open-ended* questions ("which fiscal year? please paste the 2023 and 2022 figures") rather than the protocol's yes/no questions. Manually inspecting all 173 interact trajectories, 63% terminate on an unanswered open-ended question and 35% exhaust the 10-round budget still asking; only 1 instance ever commits a value (also wrong), and just 1.3% were format-unparseable. Even in Answer+Search, where no interact channel exists, it asks instead of answering, capturing *neither* the intended nor the default reading (0.0/0.0%). This is an extreme form of Finding 12's integration gap and a distinct failure mode: clarification without commitment. The collapse is not an artifact of the yes/no protocol: with a *free-form* simulator that answers its open-ended questions from C (Appendix F), GPT-5-mini still resolves 0% across all 50 instances (zero wrong-to-correct conversions), so the failure is integrating a received clarification, not the interaction grammar.

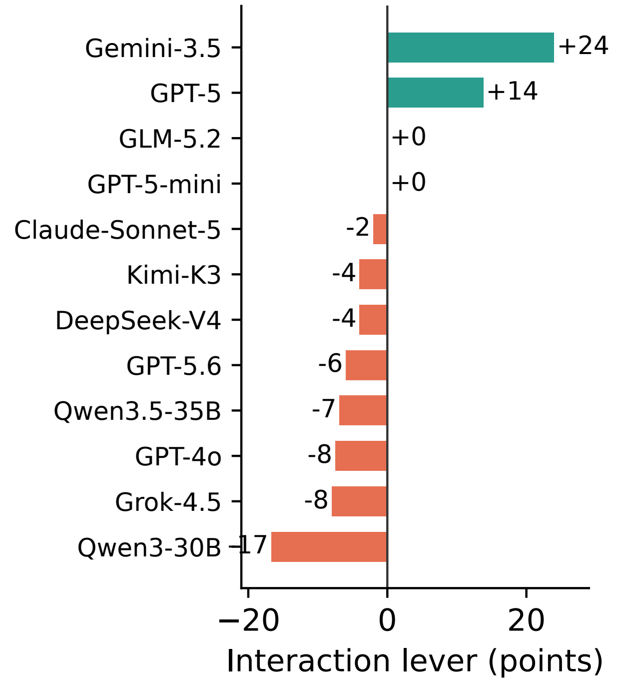

Figure 7: **Interaction is a capability, not a free lever.** Added-interaction minus plain-search accuracy across the twelve full-scale and cross-vendor models. Interaction improves accuracy for only GPT-5 and Gemini and is flat or negative for the other ten, so the ability to convert clarification into a correct answer is rare.

Language and category breakdown. Splitting by language confirms the benchmark's bilingual-difficulty hypothesis: Chinese instances are consistently harder than English in the interact mode (GPT-5 18.3% ZH vs. 24.5% EN; GPT-4o 0.8% vs. 13.2%), reflecting more complex corporate-structure naming and metric conventions in A-share filings. Stratifying by primary category (interact mode), `metric_definition` is the hardest well-populated category (GPT-5 12.7%, n=63) and `entity_scope` the next

(23.4%, n=94), while `temporal_scope` is comparatively easy (57.1%, n=7). This inverts the general-domain pattern where entity disambiguation dominates, and identifies metric-definition ambiguity as the distinctive financial challenge.

Difficulty and headroom. Per-instance accuracy falls monotonically with disambiguation entropy H_0 (from 10% at H_0=1 to under 1% at $H_0 \approx 3.6$), validating H_0 as a difficulty proxy; **no full-scale model solves 79 of 173 instances (46%) under free interaction**. These are not broken items: 96% of them become solvable under the context-oracle, so they are hard specifically at the elicitation step, not invalid or unanswerable. With the best free-interaction accuracy at 28.9% full scale (and 44% in the cross-vendor pilot), the benchmark is far from saturated. The supplementary clarification-efficiency metric mirrors the accuracy ranking in interact mode (DisE$^+$ = 0.125 for GPT-5 vs. 0.039 for GPT-4o): GPT-5 not only answers more ambiguous queries correctly but does so with fewer wasted asks.

C.3 Human Baseline

To bound the task from above under *realistic* interaction, where the disambiguating context must be *elicited*, not handed over, we collect a human baseline on a stratified 50-instance subset (32 EN / 18 ZH, balanced by category). Two finance-literate annotators (one bilingual) face the same input a model does in the search regime: the ambiguous question and a retrieved passage containing both candidate values, position-shuffled and unlabeled. Each may ask *one* yes/no question, which the experimenter answers truthfully from the withheld context C (Yes/No/IDK), before committing a final answer. This mirrors the agent's `search`→`interact`→`answer` loop, so the human row is directly comparable to the +Interact column; we grade it with the identical GPT-4o-mini grader.

Finding 15: Even humans who can ask resolve only 30%, the task is hard, not merely mis-specified for models. Annotators reach 30.0% accuracy overall (34.4% EN, 22.2% ZH), above every full-scale model (which top out at 28.9%) but far from the 93–95% context-oracle ceiling. The gap between the oracle ceiling and the human baseline is the cost of *elicitation*: knowing the resolved answer is near-trivial, but extracting the right bit through a single clarification is hard even for experts. Across models and the human baseline, targeting and resolution *dissociate*: the models with the highest AC@1 (GPT-4o and GPT-5-mini at 0.94) post the lowest accuracy, while the human, at a lower touch AC@1 of 0.72 (one focused question rather than a sweep of several categories), posts the highest (Figure 8). The annotators also never targeted the recognition-policy category either (AC@1 0 on its eight items), so the blind spot is not model-specific. The asymmetry is sharpest in Chinese, where annotators still capture the default reading 72.2% of the time at only 22.2% accuracy. Chinese instances are therefore harder to *resolve*, not harder to *recognize*, the *gui-mu/kou-fei* (parent-attributable vs. deduction-of-nonrecurring) net-income distinction is sticky even once correctly surfaced, corroborating the language-asymmetry caveat of Appendix F with a mechanism.

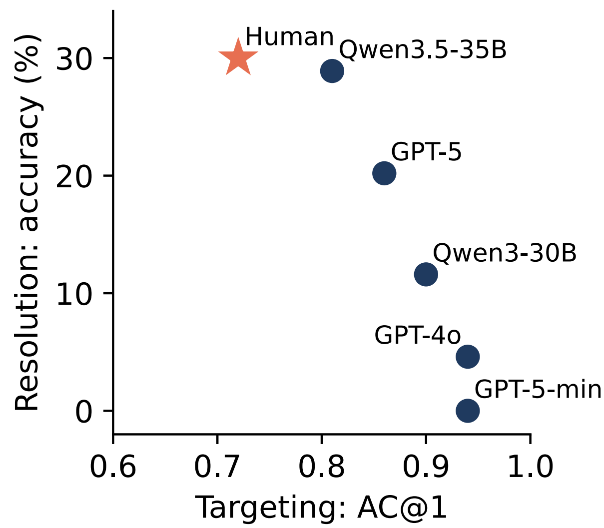

Figure 8: **Targeting does not imply resolution.** Accuracy against AC@1 (touch) in the interact regime, for the five full-scale models and the human baseline (star). The relationship is inverted: models that ask on the correct category most often resolve the ambiguity least often, whereas the human asks less precisely yet resolves more. Asking the right question and answering it are separate abilities.

C.4 Skill Dissociation Across Model Scale

Finding 16: the categories are different *kinds* of skill, some scale-solvable, one a scale-invariant blind spot. A controlled scale ladder (Qwen3 4B → 32B, Table 9) separates the categories sharply, once we look at *central* targeting. `recognition_policy` is a scale-invariant blind spot. It is never targeted at any scale, in either the touch or the central form (0.00 at 4B, 14B, and 32B, including the same 32B that targets entity well), so more parameters buy no recognition-basis targeting at all. `entity_scope` and `metric_definition` tell a subtler story. Every scale already *touches* them (touch AC@1 $\approx$ 0.9), but the ability to make them the *central* point of the question emerges only with scale, entity central-targeting rising from 0.00 at 4B and 14B to 0.68 at 32B. Scale therefore does not buy ambiguity detection for these categories, which is already present, but the precision to center the question on the right category. The dissociation confirms the taxonomy carves genuine capability boundaries rather than arbitrary strata, and it pinpoints recognition basis as the category a targeted intervention must address (§C.7).

C.5 Qualitative Case Studies

Three instances make the failure modes concrete, one for each category where the pattern is sharpest. Every case pairs a natural analyst question with two defensible readings whose ground truth is exact and machine-verifiable.

Table 9: Zero-shot *central*-targeting (central-form AC@1) by base-model scale (Qwen3, `+interact`, denominator = instances where the model asked). Under the validated touch form every scale already reaches ≈ 0.9 on entity and metric, so scale buys the precision to *center* the question on the right category rather than detection. Entity central-targeting scales steeply and is largely solved at 32B, while recognition policy stays 0 at every scale, including the same 32B that targets entity well. Recognition is underpowered ($n \leq 9$ asked-on) but flat throughout.

Category	4B	14B	32B
entity_scope	.00	.00	**.68**
metric_definition	.05	.00	.17
recognition_policy	**.00**	**.00**	**.00**

Entity scope (resolvable, yet over-elicited). *"What was Meta Platforms' operating income?"* is \$46.75B for the total consolidated company and \$62.87B for the Family of Apps segment, a 34% gap on the same GAAP metric and fiscal year. A search-only agent retrieves a passage containing both figures and commits to one, and single-gold grading credits it whenever that figure happens to match the key. This is the category models handle best, and the base policy often resolves it directly (§C.7), which is precisely why forcing a clarifying question on entity scope can *lower* accuracy rather than raise it.

Recognition policy (the blind spot). *"What were General Electric's revenues?"* is \$67.95B under GAAP revenue combining over-time service recognition with point-in-time product recognition, but \$26.79B counting only revenue recognized at a point in time, a $2.5\times$ difference driven entirely by the recognition basis. This is the category models almost never surface: instead of asking which recognition basis is meant, the typical clarification reverts to the more salient period or entity question (§4.4), so the ambiguity is never raised and the default is reported as fact. The category is represented in the network yet not acted on (§C.6). We stress that this category is exploratory ($n{=}9$) and that an item-level audit (Appendix F) finds most of its instances to be ASC 606 timing disaggregations of the kind shown here, where neither value is the total a non-expert would assume; we therefore treat the pattern as suggestive rather than quantified.

Metric definition (clarification supplies the answer). *"What was Workday's operating income growth rate?"* is 126.8% on a GAAP basis and 26% on a non-GAAP basis for the same year-over-year comparison. Here the base model cannot answer correctly without knowing which definition is intended, so a single on-category question converts a wrong answer into a right one, the mechanism behind the $+45$ point training-time gain on this category (§C.7).

Together the three cases trace the arc of the paper: a category that is resolvable but over-elicited (entity), a category that is represented yet never acted on (recognition), and a category where eliciting the definition is the whole battle (metric).

C.6 Mechanistic Case Study: Where Ambiguity Lives in the Network

The behavioral results (§4) show that frontier models default to the naive interpretation rather than clarifying. A natural question is whether models internally *represent* this ambiguity at all. Because this requires access to hidden states, we probe four open models as a proxy, spanning two size classes and two architectures: **Qwen3-4B-Instruct** (dense, 37 layers), **Qwen3.5-4B** (hybrid linear/full attention, 33 layers), and the mixture-of-experts **Qwen3-30B-A3B** (49 layers) and **Qwen3.5-35B-A3B** (41 layers).

Method. For each instance we form a within-item contrast: the bare ambiguous question Q versus Q concatenated with its disambiguating context C. We define a per-layer *ambiguity direction* as the difference in mean final-token activations between the two conditions, and score any hidden state by its (mean-centered) projection onto that direction; tracing this across layers gives a depth profile of how strongly each layer separates an under-specified question from a disambiguated one. Independently, on the two larger models we train a per-layer linear probe to decode the ambiguity *type* (entity_scope vs. metric_definition), which, unlike a context-presence detector, cannot be solved trivially.

Result. Figure 9(a) overlays the depth profiles (normalized for differing depth and scale). On all four models the signal is near-zero through most of the stack and rises sharply to a peak in the *final few layers* (≈95–98% relative depth: layers 35/37, 32/33, 48/49, 40/41). Figure 9(b) shows that the ambiguity *type* is linearly decodable at 0.80 accuracy (majority baseline 0.60) on both larger models, though it becomes available at different depths, early (≈27%) in Qwen3-30B and late (≈85%) in Qwen3.5-35B. Across four models, two scales, and three architecture families, the representation is consistent: financial-query ambiguity, and its category, is linearly encoded, emerging toward the top of the network.

Representation, elicitation, and the cost of interacting. The two MoE models we probe are also *evaluated* (Table 6): beyond linearly encoding the category, both *elicit*, and they differ in *how* they ask. Qwen3-30B interacts on 72% of instances but loops without committing in Chinese (76% of its interactive transcripts never produce a final answer), whereas Qwen3.5-35B interacts less (36%) but commits in 95% of cases and targets the gold category more often (AC@1 0.40 vs. 0.16). With the full retrieval corpus in place their accuracy is now measured, and it exposes a sharper pattern: plain search reaches 28.3%/35.8%, but *adding* interaction *lowers* accuracy to 11.6%/28.9% (-16.8/-6.9 points). Interaction is thus net-negative relative to search for both open models, as it is for GPT-4o ($12.1 \rightarrow 4.6$), and only the strongest model, GPT-5, converts clarification into a gain ($6.4 \rightarrow 20.2$). An interpretation-oracle control (the resolved interpretation supplied but no evidence) scores $\approx 0\%$ for both MoE models, so the residual gap to the oracle ceiling is an evidence-grounding cost, not a parametric-recall one. These two open models therefore do act on the represented category to the extent of *asking*, unlike the closed models, yet asking still fails to convert into a correct answer, which is the same integration gap seen throughout. A linear-probe replication on the *base* (non-instruct) checkpoints reproduces the decodability and the early-vs-late depth split (peaks 0.71/0.74 vs. baseline 0.60).

Finding 17: Open models linearly represent both ambiguity and its category, replicating from 4B to 35B and across dense and MoE architectures. The representation is type-specific (entity vs. metric decodable at 0.80) and robust to scale and architecture, yet behaviorally these same models still default rather than clarify. The behavioral gap is thus better read as a failure to *act* on a represented ambiguity than a failure to register it, though establishing this causally (e.g. via activation steering) remains future work. Concurrent general-domain work reaches the same behavioral conclusion from the outside, finding that models identify ambiguity when asked to judge it yet overwhelmingly default to direct answers (Su and Cardie 2026); our probe adds that the signal is linearly available in the representation, and our benchmark supplies the financial setting in which the cost of not acting on it is exactly measurable.

Scope and caveats. This is mechanistic evidence on open proxies, not the closed frontier models benchmarked in §4; diff-in-means probing and linear decoding establish *availability*, not causal use; and category-type decoding is supported only for the two well-populated categories (entity_scope, metric_definition). We deliberately do *not* draw behavioral conclusions from these open models' free-form generations, where surface cues (chain-of-thought markers, retrieval that surfaces the evidence span) confound naive clarification counts; the asking-behavior claims come solely from the graded closed-model evaluation in §4. A further caveat is that the ambiguity direction is a difference between Q and $Q{+}C$ inputs, which also differ in length and surface lexicon, so it may partly encode the presence of added context rather than ambiguity as such. Matched neutral-context controls, a bag-of-words lexical control, a cross-company split, and causal steering are needed before reading the probe as more than evidence that category information is *linearly decodable* from the representation. That the category is *represented but not acted on* frames the bottleneck as an action problem on an available signal, and motivates the category-conditioned interventions of §C.7, which target exactly that signal.

C.7 Diagnosis and Intervention: Acting on the Ambiguity Category

Our results converge on a single diagnosis. The benchmark localizes the bottleneck to *eliciting* the disambiguation: agents answer 93–95% correctly once the intended interpretation is supplied (§4.2) but far less on their own (a mean of 8% across the OpenAI ladder), and the mechanistic probe shows why, open models *linearly represent* the ambiguity *category* yet do not act on it (§C.6). The remaining gap is thus an *action* problem on an already-available signal, and the *category* is its natural unit. We now turn that diagnosis into intervention: we first catalog how acting fails (a seven-type error taxonomy), then present two interventions that share one idea, **condition on the category**, instantiated at inference time (Category-Aware ReAct, §C.7) and at training time (category-guided SFT/GRPO, §C.7).

Error Taxonomy Beyond aggregate accuracy, we characterize model failures using a seven-type error taxonomy (Table 10) derived from FinInteract's evaluation signals. Each type is diagnosable from existing metrics without additional annotation.

Table 10: Error taxonomy E1–E7 with diagnostic signals and improvement directions.

Type	Description	Improvement direction
E1	**Ambiguity blindness**: no interaction, wrong answer	Ambiguity pre-check; train on detection
E2	**Wrong-category clarification**: asked wrong dimension	Category classifier; financial taxonomy grounding
E3	**Generic clarification**: vague ask, no category target	Category-conditioned clarification templates
E4	**Retrieval dominance**: searched but skipped interaction	Force interpretation check after retrieval
E5	**Clarification misuse**: right question, wrong answer	State-gated answering; stronger grounding
E6	**Evidence grounding**: template-oracle correct, agent wrong	Improved source-grounded extraction
E7	**Over-interaction**: correct but low DisE$^+$	Penalize via DisE$^+$/AC@1 efficiency signal

Diagnostic Mapping from Evaluation Signals The existing evaluation metrics map directly onto error types, enabling failure attribution without additional annotation:

- **Low IR + low Acc** $\rightarrow$ E1 (ambiguity blindness dominant)
- **High IR + low AC@1** $\rightarrow$ E2/E3 (interacts but misdirected)
- **High AC@1 + low Acc** $\rightarrow$ E5/E6 (right category, wrong grounding)
- **High GenericAskRate** $\rightarrow$ E3 (lacks financial clarification structure)
- **High WrongCategoryRate** $\rightarrow$ E2 (confidently misdirected)
- **Always-ask** $\gg$ **Standard ReAct** $\rightarrow$ E1 dominant; latent capacity exists
- **Template-oracle** $\gg$ **Agent-interact** $\rightarrow$ "what to ask" is the bottleneck
- **Category-oracle** $\gg$ **Agent-interact** $\rightarrow$ category recognition is the bottleneck
- **Template-oracle still low** $\rightarrow$ E6 residual; evidence extraction still fails

Misintegration, not misdirection, dominates the error budget. Attributing each interact-mode outcome to the taxonomy (Figure 10) shows that *E5* (right category touched, wrong answer) accounts for the large majority of failures, 67% for GPT-5, 89% for GPT-4o, 90% for GPT-5-mini,

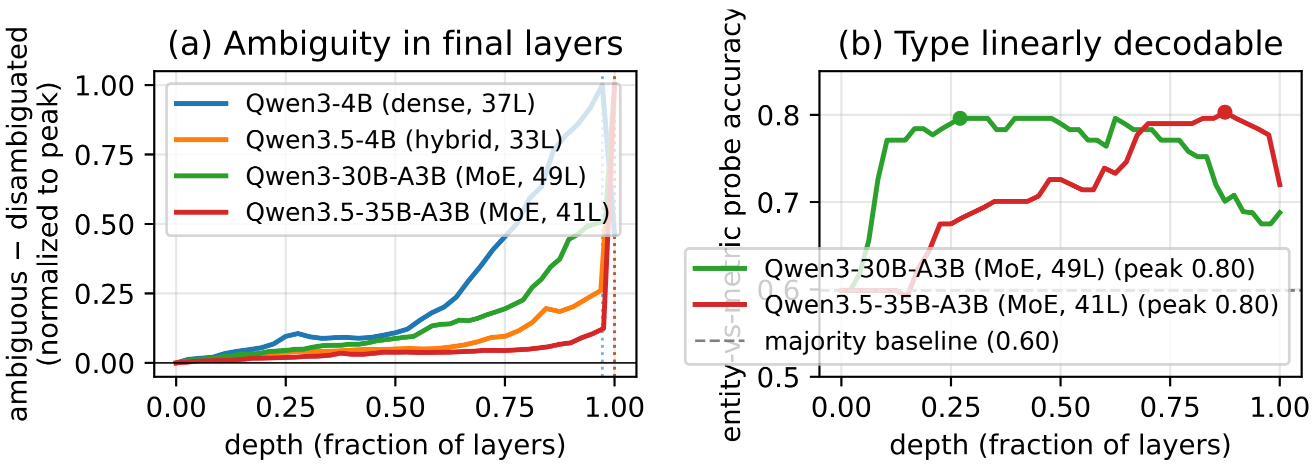

Figure 9: Four-model depth profile of the ambiguity signal (Qwen3 4B→35B, dense and MoE). (a) Ambiguous−disambiguated separation rises to a peak in the final layers of every model (dotted verticals mark per-model peaks; normalized for depth and magnitude). (b) The ambiguity *type* (entity vs. metric) is linearly decodable at 0.80 on both MoE models, well above the 0.60 majority baseline, peaking at different depths.

and 75% for Qwen3-30B, while not-asking (E1) and off-category asks (E2 and E3) together stay below 15% for the closed models. The bottleneck is converting a correctly targeted clarification into the intended answer, not deciding whether or what to ask. Qwen3.5-35B-A3B is the informative exception, asking less (E1 21%) but resolving more of what it attempts (Correct 35%), so its failures lean toward under-asking rather than misintegration. This corroborates Findings 7 and 12 across the full error budget and motivates the state-gated answering of §C.7.

Inference-Time Intervention: Category-Aware Clarification ReAct The first intervention conditions on the category *without changing the model weights*. Standard ReAct leaves the agent free to decide whether and what to ask. We propose **Category-Aware Clarification ReAct**, a three-component structured interaction policy that directly addresses E1–E3.

Component 1: Ambiguity Pre-Check. Before any action, the agent explicitly decides which of the five ambiguity categories are underspecified in the question and initializes an interpretation state $\mathcal{S} = \{\text{entity, period, metric, basis, vintage}\}$, marking each field as *resolved* or *unknown*.

Component 2: Category-Conditioned Clarification. For each *unknown* field, the agent generates a targeted yes/no question using category-conditioned templates (one per category) rather than free-form clarification. The agent emits at most one clarification per turn, prioritizing the most information-theoretically valuable unknown field.

Component 3: Interpretation-State-Gated Answering. The agent may only emit a final answer once all required fields are resolved (or the user explicitly declines to clarify). This prevents the *retrieval dominance* pattern (E4) where the agent answers immediately after finding a plausible number in a search result.

Expected outcomes. Relative to standard Answer+Search+Interact, Category-Aware ReAct should reduce GenericAskRate (E3), WrongCategoryRate (E2), and IR=0 failures (E1), while improving AC@1 and ultimately Accuracy. Template-oracle provides an upper bound: if Category-Aware ReAct approaches template-oracle performance, it confirms that category recognition is the dominant remaining bottleneck after the E1–E3 errors are addressed.

Pilot results. We evaluate the policy on a GPT-5 backbone over a stratified $N{=}50$ subset run through the same OpenRouter harness, alongside four matched baselines (Standard ReAct, Always-Ask, Category-Oracle, Template-Oracle). Category-Aware ReAct reaches 46.0% accuracy against the 34.0% of Standard ReAct in the same run, a $+12$ point gain, and posts the best calibration of any condition (ECE 0.241 versus 0.308 for Standard ReAct). Always-Ask collapses to 10.0%, confirming that asking indiscriminately is worse than not asking and that the gain comes from asking *selectively* on the right category. The single-category diagnostics (AC@1, WrongCategoryRate) are not informative for this policy: because the Ambiguity Pre-Check deliberately enumerates every candidate category to rule each in or out, an any-category touch form saturates near 1.0 while the strict central-category form is near 0, so neither cleanly credits the policy and we omit AC@1 for it in Table 6. The accuracy and calibration gains are the load-bearing result. We report this as a pilot rather than a full-panel row because it is a single backbone at $N{=}50$, and we expect the effect size to tighten at full scale. One caveat bears directly on the mechanism: the policy bundles all three components above, and we do not ablate them here, so part of the $+12$ may come from the state-gated commit (Component 3, an integration fix) rather than from better category-conditioned elicitation (Component 2). A generic-structured clarification baseline that keeps the structure but

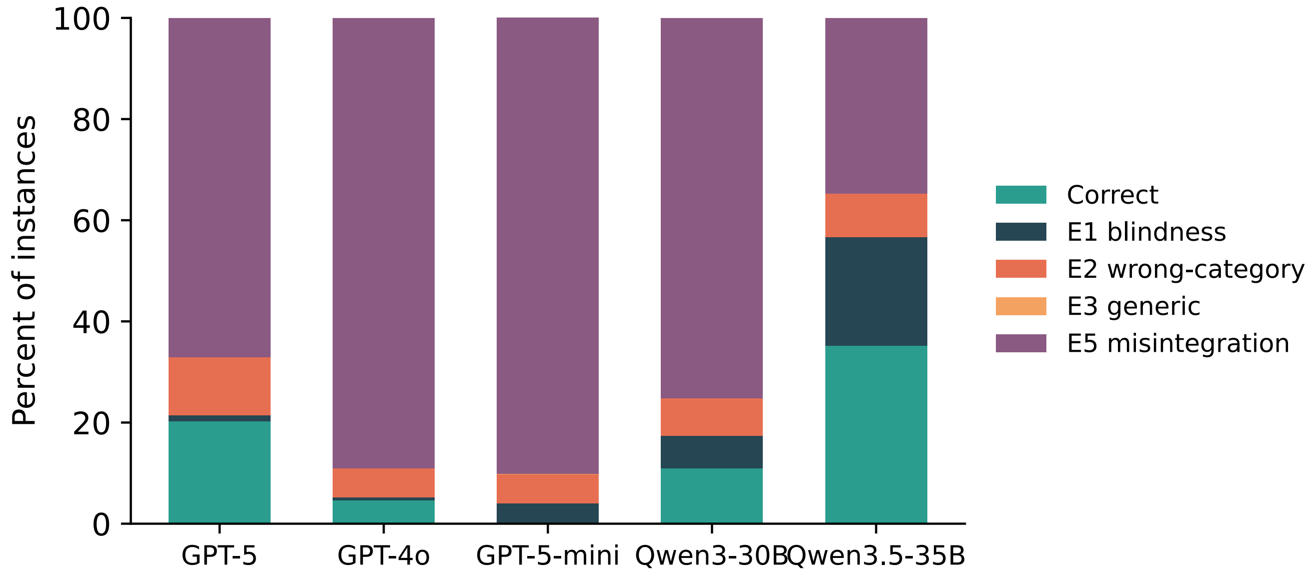

Figure 10: **Error-type distribution** in `+interact`, attributing every instance to the taxonomy of Table 10 with the corrected category classifier. *E5 misintegration* (the agent asks, touches the correct category, yet answers wrong) dominates for every model, while ambiguity blindness (E1), wrong-category (E2), and generic (E3) asks stay comparatively rare.

drops the named categories, together with a per-component ablation, is the clean control and remains future work.

Training-Time Intervention: Category-Guided SFT and GRPO The per-category skill analysis (§4.4) gives this intervention a precise target: models do not lack clarification ability uniformly, they lack *specific* category skills (AC@1 = 0 on recognition policy, partial on metric). The question is whether those missing skills can be installed by training rather than prompting. The second intervention applies the *same* idea, condition on the category, but in the weights rather than the prompt, showing the benchmark is not only diagnosable but *trainable*. Starting from Qwen3-4B-Instruct (4-bit QLoRA), we run an RLVR ladder, supervised fine-tuning (SFT), KTO, then GRPO, on the FinInteract training split, with all user simulators, graders, and category judges served locally (zero API cost). We score each stage with the two *leak-proof* metrics, AC@1 and interaction rate (Table 7); accuracy is reported only with the caveat that the training environment's retrieval returns the disambiguated evidence span, so it is partly leaked and is *not* a reliable learning signal.

The category-guided teacher. The decisive ingredient is how SFT demonstrations are generated. An uninformed teacher, even a 72B model, almost always asks the *obvious* ambiguity (typically the reporting period) and rarely the subtle gold category (entity scope, recognition policy), producing $\sim$0% on-category demonstrations. Privately revealing the gold category to the teacher while generating demonstrations (the hint is *not* stored in the trajectory) lifts the on-category yield to $\sim$79%, a reusable recipe for any category-structured clarification task.

SFT does the heavy lifting. The leak-proof signals jump almost entirely at the supervised stage, AC@1 rises $25 \rightarrow 69$ and interaction $55 \rightarrow 100$ (Figure 3a), while KTO and GRPO move within run-to-run noise at n=51. Training teaches the model to *always interact and target the correct category*, the behavior FinInteract is built to elicit. Whether this gain reflects a sharpened internal *representation* of the category or only a rewired output *policy*, probing the trained model as in §C.6, is a direct test of the diagnosis and a target for future work.

True multi-turn GRPO. The ladder above optimizes a single-turn approximation. We additionally confirm the full multi-turn episode is trainable end-to-end: optimizing the 4B policy over entire ReAct episodes (loss masked on simulator and retrieval tokens) for 15 steps raises mean reward $0.55 \rightarrow 1.08$ while clarifying turns rise $5.7 \rightarrow 8.6$ (Figure 3b), the category-aware reward is teaching the policy to ask more. We report this as a learning-signal demonstration, not a converged model.

Caveats. Results are on a held-out split of n=51; all judges and the teacher are local Qwen2.5-32B/72B models (weaker than GPT-4o, and siblings of the policy); the terminal reward is partly leaked; and GRPO is run for tens of steps, not to convergence. We therefore frame this as evidence that FinInteract is *trainable*, a category-guided SFT seed instills category-targeted clarification, rather than as a state-of-the-art accuracy result.

When Can the Benchmark Refresh Itself? Three Co-Evolution Probes A benchmark that models eventually memorize stops measuring. Because FinInteract couples a grounded data-generation pipeline (§3.3) to a trainable policy (§C.7), it can in principle *co-evolve*: generate fresh instances on a category the current model fails, train on them, and repeat, a data$\leftrightarrow$model loop in the spirit of selfimproving generators (Chen et al. 2026), but *grounded* in real filings rather than the ungrounded self-play those methods risk. We run three gated rounds, on the scarce, subordinate category (`recognition_policy`), the abundant, salient category (`entity_scope`), and the abundant-but-only-weakly-scale-learnable category (`metric_definition`, §4.4), to ask not *whether* the loop works but *when*.

All three rounds run end-to-end: the Constructor→Verifier pipeline yields hard instances (every one passes the adversarial gate: the base fails it answer-only) and category-guided SFT converges. The outcome splits cleanly by *abundance*, not by scale-learnability. On both abundant categories, `entity_scope` and `metric_definition`, Targeting generalizes to held-out *human* items (.00 → .68 and .00 → .58, each ∼4× the +.15 bar) and to held-out *generated* items, with no catastrophic forgetting. On the scarce `recognition_policy` it does not move (.00 on both held-out sets): the trained model reverts to the salient period/entity question rather than the recognition-basis distinction. The contrast rules out a broken pipeline, the *same* machinery installs entity and metric but not recognition, and, because metric is only *weakly* scale-learnable yet trains as well as entity, it rules out scale-learnability as the gate (§C.7).

Finding 18: the loop closes on abundant categories and fails on scarce ones; whether elicitation *helps* is a separate, per-category property. A self-refreshing FinInteract is viable wherever the data arm can supply enough frontier instances, both abundant categories install and generalize; the scarce category (recognition's 36-instance ceiling from just $30/1{,}847$ primary passages) does not reliably (Finding 21). But installing on-category elicitation is not automatically beneficial: it *helps* metric (accuracy 15 → 60, +45: the base cannot answer without the metric definition, so the clarifying question *supplies* it: 18 wrong→correct, 0 regressions) and *hurts* entity (accuracy 32.5 → 22.5, −10: the base can often answer the entity directly, so forcing a question turns six correct direct answers into wrong multi-turn ones). Both drove interaction to 100%; the accuracy sign flips on whether the base could already answer, the training-time face of the "interaction can hurt versus plain answering" effect seen at inference (§4.2). The actionable step, which we demonstrate next, is a *conditional when-to-ask gate*, elicit only where the model cannot already answer, plus multi-round curriculum and fresh-filing ingestion to lift the scarce categories above their availability ceiling.

Finding 19: the over-elicitation regression is fixable at inference, but not by a train-time gate. We test the when-to-ask gate directly on the frozen human probes (n=40 per category). A *train-time* gate that mixes no-ask demonstrations into the category-guided SFT backfires: even 7 of 164 no-ask demonstrations tip the policy into a bimodal collapse in which it answers directly without retrieving, which is fatal because in FinInteract almost nothing is answerable from parametric memory (entity accuracy 32.5 → 7.5, metric 60 → 2.5). An *inference-time* gate recovers the loss with no extra training. Routing by the base model's own ask decision, taking the co-evolved asker's answer when the base chose to ask and the base's direct answer otherwise, lifts entity back to 30.0 from the always-ask 22.5 (matching the base's own 32.5) at a 40% ask rate rather than 100%, and keeps +20 of metric's +45 gain (35.0 versus base 15.0). Over-elicitation is thus an inference-time routing problem rather than a training one. The residual metric gap traces to the base's uncertainty signal under-asking where metric needs a question, so a confidence-calibrated gate that asks when the silent answer would be *wrong* rather than when the base is merely unsure is the concrete next lever.

What Gates Co-Evolvability? A Falsification Test The three rounds let us isolate *which* category property predicts whether the loop closes. Two candidates are natural, a category's *scale-learnability* (does zero-shot Targeting rise with model size?) and its *source-abundance* (can the data arm supply frontier instances?), and we had also hypothesized a representational version: *learnable iff the category is linearly decodable* from hidden states. The `metric_definition` round was designed to discriminate the first two; a linear probe tests the third. Table 12 collects the three candidate predictors against the observed outcome for every category. Both tidy hypotheses fail.

Scale-learnability does not gate co-evolution. `metric_definition` is abundant (498 primary passages) but only *weakly* scale-learnable (zero-shot AC@1 .05 → .26, slope +0.21, versus entity's +0.77). We pre-registered the two-factor prediction that it would co-evolve only *partially*. It did not: metric is a *full* GO (Targeting +0.55, matching entity's +0.61; Table 11). What separates GO from NO-GO is not the scale-slope but *abundance*, both abundant categories close the loop; only the scarce category fails.

Representation does not gate it either, and is not the bottleneck. A per-layer one-vs-rest linear probe decodes every category from base hidden states, and the pattern *anti*-tracks learnability: `recognition_policy`, the unlearnable, NO-GO category, is the *most* decodable (AUROC 0.93 on a size-matched sample, above entity/metric at ∼ 0.77); decodability is *flat* across the 4B → 32B sweep even as behavioral AC@1 climbs .12 → .89; and category-guided SFT leaves it unchanged. The category is fully *represented* at every scale, before and after training: the model simply does not *act* on it, the "represents-but-doesn't-act" gap of §C.6, now shown to hold even for the category co-evolution cannot fix.

Finding 20: co-evolvability is gated by source *abundance*, driven by *salience*, not by scale-learnability or representational decodability. What survives both falsifications is a single upstream cause. `recognition_policy` is a candidate category in 796 passages but *primary* in only 30 (salience 0.04): it is almost always the *subordinate* ambiguity, behind a more salient entity or period category in the same passage. Low salience produces both the scarcity that starves the data arm and the salient competitor the model's prior reaches for instead, so the representation is present but never acted on and cannot be cheaply trained in. Salient categories behave oppositely (entity, salience 0.92: abundant and trainable). For a practitioner, **salience, not scale curves or probe accuracy, is the cheap, up-front signal** of which categories a co-evolving benchmark can keep hard and which

Table 11: Three co-evolution rounds (category-guided SFT on a 4B policy). Targeting (AC@1) is on a *held-out human* probe never seen in training. The loop closes on both *abundant* categories and fails only on the *scarce* one; whether asking *helps accuracy* flips on whether the base could already answer.

	`entity` abundant	`metric` abundant	`recognition` scarce
Frontier instances generable	99	76	36 (ceiling)
Targeting, base → evolved	.00 → **.68**	.00 → **.58**	.00 → .00
on held-out *generated*	.40	.88	.00
Interaction rate	40 → 100	50 → 100	67 → 100
De-leaked accuracy	32.5 → 22.5	15 → **60**	11 → 11

Table 12: What predicts co-evolvability. The observed outcomes track *abundance/salience*, *not* the AC@1 scale-slope (metric: low slope, GO) and *not* decodability (recognition: highest AUROC, yet does not co-evolve). Salience = primary/any-candidate passages; decodability = peak one-vs-rest AUROC (recognition on a size-matched augmented sample; raw n=9 overfits to 1.0). *metric is a GO but needs ∼2 accumulated slices (∼50 instances); one 30-instance slice teaches it nothing (Finding 21). ‡recognition is *unreliable* rather than a clean fail: held-out human AC@1 ranges 0–0.44 across training runs on the scarce n=9 probe.

Category	Primary	Salience	Slope	Decode	Co-evolve
entity_scope	905	**0.92**	+0.77	0.77	**GO**
metric_definition	498	0.48	+0.21	0.77	**GO***
temporal_scope	150	0.15	sat.	0.92	solved
filing_vintage	264	0.33	–	–	untestable
recognition_policy	30	**0.04**	+0.00	**0.93**	weak‡

need fresh human curation. The corollary for evaluation is sharper: the hardest ambiguity to make models *handle* is not the one they cannot represent, but the one that is always someone else's more-obvious question.

Finding 21: salience predicts a category's *data-need*, not just a binary outcome, and a validation-guarded arms race confirms it. Extending each category to a multi-round curriculum (Fig. 11), with checkpoints selected on a frozen validation set and reported on the held-out human probe, refines the binary gate into a gradient that tracks salience. **Entity** (salience 0.92) learns category-targeting from a *single* 30-instance slice and transfers to human ambiguity at 0.88–0.93. **Metric** (0.48) needs *accumulation*: one slice yields 0.0, two accumulated slices reach 0.75, so its earlier single-round GO used ∼60 instances precisely because that is roughly what the category requires. **Recognition** (0.04) is *unreliable*: across training runs its human AC@1 ranges only 0–0.44 on the scarce n=9 probe, weak and inconsistent, not a clean zero. Scoring the same runs as an *Elo arms race* between the solver and a hard-example miner (win = AC@1, which, unlike recall-walled resolution, varies across checkpoints) makes the instrument non-degenerate: entity is *solver-runaway* (solver Elo 1000 → 1551, outpacing a corpus-bounded miner that cannot synthesize adversarially harder targeting), while recognition *stalls* (flat). Two guarded rounds peak; the third overfits and is rolled back. The practical reading: salience is an up-front estimate not just of *whether* an category co-evolves but of *how much* data it needs, and one well-targeted round, guarded on held-out validation, is close to the sweet spot.

D Extended Related Work

This appendix expands the related work summarized in §6.

Financial QA Benchmarks. Table 13 compares FinInteract to major financial QA benchmarks. FinanceBench (Islam et al. 2023), FinQA (Chen et al. 2021), DocFinQA (Reddy et al. 2024), BizBench (Krumdick et al. 2024), FinBen (Xie et al. 2024), and FinanceQA (Mateega, Georgescu, and Tang 2025) all enforce single-gold-answer conventions and evaluate models in a single turn; FinanceQA further shows that frontier models fail ∼60% of realistic analyst tasks, but attributes this to numerical rigor rather than ambiguity. FinEval (Guo et al. 2025), a large Chinese financial-knowledge benchmark, incidentally identifies an "ambiguity handling weakness" error type in its agent tasks but does not isolate or control it. Recent *agentic* financial benchmarks raise task realism, Finance Agent Benchmark (Bigeard et al. 2025) (537 expert research tasks, where the best model reaches only 46.8%) and FinAgentBench (Choi et al. 2025) (agentic retrieval over EDGAR), but both remain single-turn and single-gold, evaluating analysis and retrieval rather than whether an agent recognizes that a query is under-specified. FinInteract is the first to make query ambiguity the central, controlled construct, via paired default/intended interpretations and multi-turn interaction, in the financial domain.

Ambiguous QA. AmbigQA (Min et al. 2020), ASQA (Stelmakh et al. 2022), CondAmbigQA (Li et al. 2025), and CLAM (Kuhn, Gal, and Farquhar 2023) have matured in the general (Wikipedia) domain. More recent benchmarks target the agent's *own* ability to ask: CLAMBER (Zhang et al. 2024) evaluates whether models

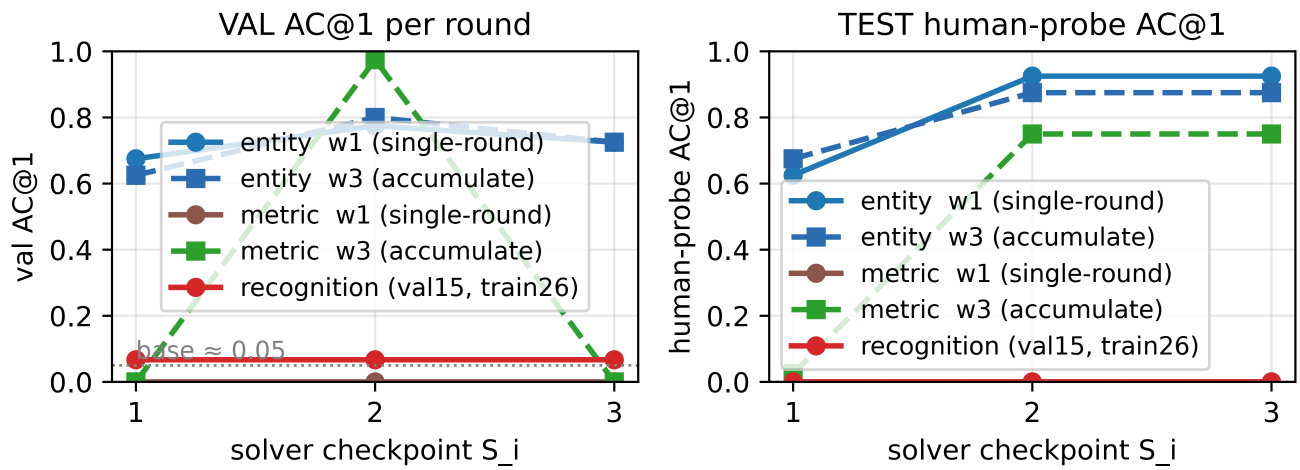

Figure 11: Multi-round co-evolution with a validation guard. *Left*: validation AC@1 (selection metric) per round. *Right*: held-out *human*-probe AC@1 (never used for selection). Entity learns targeting from a single 30-instance slice and transfers to humans ($0.63 \rightarrow 0.93$); metric needs *accumulation* (one slice teaches nothing, brown, flat, while two accumulated slices reach 0.75, green); recognition stays at base (≈ 0.05) even with a fair 26-instance set. The guard preserves human transfer (it rolls back the round-3 candidate, which regresses on validation).

identify and clarify ambiguous information needs, and QuestBench (Li, Kim, and Wang 2026) tests whether they can ask the minimal necessary question in reasoning tasks. Closest in spirit to our diagnosis, Su and Cardie (2026) show in the general domain that models recognize ambiguity when asked to judge it yet rarely ask about it in ordinary question answering, and that retrieved context makes them ask *less*, a result our retrieval-augmented rows reproduce in finance. These works are general-domain and judge a single clarifying step; FinInteract adds *per-category* targeting (AC@1), programmatically verifiable financial grounding, and, via the co-evolution rounds (§C.7), an analysis of *when* asking helps versus hurts. We adapt this line's formalism to finance, replacing human-judged ambiguity labels with programmatically verifiable XBRL-grounded evidence pairs.

Clarifying Questions. Asking clarifying questions is a long-standing goal in information retrieval and dialogue. Rao and Daumé III (Rao and Daumé III 2018) rank clarification questions by expected value of perfect information; ClariQ (Aliannejadi et al. 2020) established a shared task for generating clarifying questions in open-domain dialogue; and the mixed-initiative paradigm is surveyed in Conversational Information Seeking (Dalton et al. 2022). FinInteract instantiates this lineage with a leak-proof, per-category measure of *which* clarification an agent should issue and a verifiable outcome for whether issuing it resolves the query.

Interactive Search Agents. InteractComp (Deng et al. 2026) is the direct methodological ancestor. We extend its general-domain setting to finance with a domain-specific ambiguity taxonomy and bilingual evaluation. BrowseComp (Wei et al. 2025) tracks longitudinal retrieval trends; we test whether the interaction-capability gap identified there persists in finance.

E Extended Discussion

This appendix expands the discussion points summarized in §5.

Elicitation is a trainable skill that current models lack. Our central result is that models fail to resolve ambiguity not for want of knowledge but for want of asking the right question (RQ2). This reframes the problem for future work. Progress will come less from larger backbones, which barely move interaction accuracy, and more from policies and objectives that reward asking on the correct category at the right time. The Category-Aware ReAct policy and category-guided fine-tuning (RQ3) are first steps, and both leave a large gap to the oracle ceiling, so reward shaping on the ambiguity category and calibrated deferral are natural next directions.

Categories are different learning problems. The per-category dissociation shows that some clarification abilities are solved by scale while others, in particular the recognition basis, resist it and are only unreliably teachable. A single aggregate interaction score therefore hides where the real difficulty sits. Future benchmarks and training curricula should treat each category as its own learning problem and budget data accordingly, which our salience estimate begins to quantify.

Human and AI collaboration under ambiguity. Because models confidently return the default reading, the safe behavior is often to ask rather than to answer. This motivates evaluation and interfaces that credit well-timed clarification and calibrated abstention rather than raw commitment, and

Table 13: Comparison of financial QA benchmarks. ✓= supported; – = not applicable or not evaluated.

Benchmark	Scale	Ambig.	Interact.	Multi-turn	Biling.	Evidence	Grounding
FinanceBench (Islam et al. 2023)	150	–	–	–	–	human	SEC
FinQA (Chen et al. 2021)	8,281	–	–	–	–	human	10-K/Q
DocFinQA (Reddy et al. 2024)	7,437	–	–	–	–	human	10-K/Q
BizBench (Krumdick et al. 2024)	multi-task	–	–	–	–	auto	mixed
FinBen (Xie et al. 2024)	multi-task	partial	–	–	partial	auto	mixed
FinanceQA (Mateega, Georgescu, and Tang 2025)	multi-task	–	–	–	–	expert	SEC
Finance Agent Bench. (Bigeard et al. 2025)	537	–	–	–	–	expert	SEC
FinAgentBench (Choi et al. 2025)	26K	–	–	–	–	auto	SEC
FinEval (Guo et al. 2025)	8,351	–	–	–	ZH	expert	exams
FinInteract (ours)	173	✓	✓	✓	EN+ZH	LLM+human	EDGAR+CNINFO

it connects to mixed-initiative retrieval where the cost of a confident wrong answer is high.

Self-refreshing benchmarks and broader scope. Our co-evolution probes suggest a benchmark can extend itself on categories whose evidence is abundant, guarded by held-out validation, which offers a path to keep the instrument fresh as models improve. Natural extensions include more ambiguity categories, additional languages and filing regimes, and transfer of the paired default and intended construction to other high-stakes domains where a single gold answer hides genuine ambiguity.

F Extended Limitations

This appendix expands each limitation summarized in §5.

Scale. At 173 instances FinInteract is small (comparable to FinanceBench's 150 (Islam et al. 2023)); we prioritize per-instance construction rigor (every item is adversarially verified and R14-discriminating) over raw size, and Appendix G shows the pipeline scales at $0.31/instance.

Language balance. The English split (53) is smaller than the Chinese split (120); per-language results (§4) should be read with this asymmetry in mind, and the EN portion is a target for growth.

Category coverage and recognition-policy validity. The taxonomy is intentionally five-category, but `filing_vintage` is unrepresented and `temporal_scope`/`recognition_policy` are sparse (Table 4): viable structured ambiguity for these categories is rare in public filings (Appendix G). We therefore treat entity scope (n=94) and metric definition (n=63) as the two well-powered categories and the other three as exploratory. An item-level audit of the nine `recognition_policy` instances is a further caution: most are ASC 606 timing disaggregations (revenue recognized point-in-time versus over-time), where neither value is the total a non-expert would assume, and one pairs total GAAP revenue against an operating cash-flow proxy that is not a defensible reading of "revenue." Even our human annotators never targeted the category (AC@1 0 on its items). We therefore report the recognition-policy blind spot as a preliminary observation on a small, imperfect sample, not a headline result, and recommend rebuilding this category with analyst-audited pairs before drawing quantitative conclusions.

Model coverage. We report the OpenAI tier ladder (3 models) and two open MoE models at full scale (n=173); a seven-model, six-vendor panel of current frontier models is additionally evaluated in a preliminary pilot (N=50/mode, Table 6, panel (b)), which reproduces the central finding but at reduced statistical resolution, and a full-scale multi-vendor evaluation remains in progress. The human baseline (N=50, a single yes/no clarification) is a constrained reference rather than an upper bound, since models given up to ten rounds can exceed it, and it is collected from two annotators on a subset. A larger expert panel remains future work.

Grader and simulator. Correctness and category-targeting are judged by GPT-4o-mini and the user is simulated by GPT-5. We validate these against humans on a stratified, blindly double-annotated sample (five annotators). The correctness grader agrees with human consensus at Cohen κ=0.85 (raw agreement 0.92), exceeding inter-annotator agreement (0.86), so accuracy figures hold under human labels. The AC@1 classifier initially agreed only weakly (0.53). A single-label design over-assigned `temporal_scope` to compound questions that also name an entity or metric. The corrected multi-label *touch* classifier reaches 0.75 agreement, matching inter-annotator agreement (0.73), and the AC@1 values reported here use it. Because this shows models mostly do ask on-category, the corrected reading is that the gap lies in central targeting and integration rather than detection. A strict central-category variant is discriminative but not yet human-validated, so we report it only as a supplementary decomposition. Interaction results are also robust to the simulator: on a 50-instance subset, standard +Interact accuracy is $34/28/26\%$ under the LLM, deterministic, and 15%-noisy simulators. Crucially, the yes/no action grammar does not manufacture the weak-model collapse: with a *free-form* simulator that answers open-ended clarifying questions from C (for example "which fiscal year?") rather than only yes/no, GPT-5-mini stays at 0% across all 50 instances (identical to the yes/no simulator, zero wrong-to-correct conversions, three-round budget). Its failure is therefore a post-clarification integration failure, not an artifact of the interaction protocol.

Construction circularity and synthetic intent. The constructor, adversarial verifier, and user simulator are all GPT-5-family models, and the verifier removes instances the GPT-5 family answers correctly without C. This depresses GPT-5-family answer-only accuracy (0.6%) by construction relative to the non-OpenAI vendors the filter does not target (mean 10.0% answer-only across six vendors), so cross-vendor answer-only comparisons are not fully apples-to-apples and the GPT-5-family floor is a filter-induced lower bound. The elicitation gap itself is unaffected: the filter targets answer-only solvability, not interactive resolution, and every vendor still resolves far below its oracle ceiling. Verifying a subsample with a non-OpenAI adversarial ensemble is a natural robustness check for future work. A related construct-validity caveat is that questions are reverse-constructed from filings rather than drawn from real user queries, and the "intended" interpretation is a constructor-assigned hidden context. The benchmark therefore measures a model's ability to recover a *specified* interpretation, a necessary and controllable proxy for, but not identical to, disambiguating the intent of a real analyst.

Contamination. Instances are freshly constructed from FY2022–2025 structured facts and answer-only accuracy is $\approx 0.6\%$ across all sources, but we do not claim absolute contamination safety for the DocFinQA-derived minority.

All instances are ambiguous by construction. Every one of the 173 items is built to require clarification, so a model can succeed by assuming ambiguity is always present. FinInteract therefore measures *what* to ask and *how* to integrate the answer, not a calibrated *when*-to-ask policy: it does not score false-positive clarification, over-asking on already-specified questions, or the correct choice to answer directly when a mild ambiguity does not change the value. Adding matched unambiguous and answer-invariant controls is the single most important extension for turning the benchmark into a test of calibrated interaction, and we accordingly frame our claims around asking-and-integrating rather than when-to-ask.

Intervention evidence is preliminary and not component-attributed. The Category-Aware ReAct gain ($34 \rightarrow 46\%$) is a single GPT-5 backbone on $N{=}50$ (about six additional items), without a significance test, and its matched-baseline subset scores above the full-scale GPT-5 row, so the effect must be confirmed at full scale with confidence intervals. Just as important, the policy bundles three components (ambiguity pre-check, category-conditioned questioning, and interpretation-state-gated answering) that we do not ablate: the improvement may stem substantially from the state-gated commit, an *integration* fix, rather than from better category *elicitation*, so the pilot does not by itself isolate the taxonomy's mechanistic role. A generic-structured clarification baseline and a per-component ablation are the needed controls. The training-time results are likewise early, resting on a small held-out probe, gains dominated by supervised fine-tuning with only a short reinforcement phase, an optimistic oracle-retrieval reference, and interaction driven to near-total compliance rather than a learned when-to-ask policy.

A single protocol confounds ability with interface adherence. All vendors share one ReAct prompt and yes/no action grammar. GPT-5-mini's 0% is partly a protocol-adherence failure (it keeps asking open-ended questions, exhausts the round budget, and rarely commits a final answer) rather than proof of zero financial-ambiguity ability. A free-form simulator rules out the yes/no grammar as the cause (the collapse persists), but tool schema, few-shot examples, constrained decoding, and model-specific adapters all shape protocol compliance, so a fixed cross-vendor prompt is a conservative rather than a maximally fair comparison; per-model interface tuning is future work.

Scope. The paper spans a benchmark, the single-gold illusion, the taxonomy, and several exploratory studies (representation probing, inference and training interventions, co-evolution, and salience). The three main-text findings are the load-bearing contribution and the rest are explicitly exploratory and appendix-scoped; the two highest-value additions they point to, matched ambiguous/unambiguous evaluation and a component ablation of the intervention, are our priority follow-ups.

G Curation Cost to Replicate the Benchmark

A practical concern for any LLM-assisted benchmark is the cost to reproduce or scale it. We instrument the construction pipeline (§3.3) and report measured API costs. All figures use list prices at construction time (per 1M tokens: GPT-5 \$1.25 in / \$10.00 out; GPT-5-mini \$0.25 / \$2.00; GPT-4o-mini \$0.15 / \$0.60) and count API fees only, they exclude the one-time filing-collection compute and the human spot-check labor (10–20$\%$ of accepted instances).

The constructor, not the verifier, is the yield bottleneck. Re-running the pipeline over a held-out sample of candidate passages reproduces a sharp funnel (Table 14). Only 11.5% of keyword-tagged candidate passages survive the constructor, which rejects the rest as containing *no viable ambiguity* for the target category; but of those that survive, 98.5% pass the 10-trial adversarial verifier. In other words, the construction prompt is strict and the verifier rarely needs to fire, the cost of the benchmark is dominated by constructor calls spent on candidate passages that turn out not to support a genuine ambiguity, not by verification.

Projected cost to scale. At the measured blended rate, producing N accepted instances requires $\approx N/0.113$ candidate

Table 14: Measured curation cost. Unit costs are from instrumented pipeline runs (constructor-only n=20; full pipeline n=6); the acceptance funnel is from the full construction run (n=1127 candidate passages). A full pipeline pass is one GPT-5 constructor call plus ten verifier trials (five GPT-5, five GPT-5-mini) with GPT-4o-mini grading.

Quantity	Value
Constructor viability (viable / candidate)	11.5%
Verifier acceptance (accepted / viable)	98.5%
Overall acceptance (accepted / candidate)	11.3%
Cost per rejected candidate (constructor only)	$0.020
Cost per viable passage (full pipeline)	$0.155
Blended cost per candidate passage	$0.035
Cost per accepted instance	**$0.31**

passages and the following API spend: 173 instances (the current release) ≈ $54; 500 instances ≈ $155; 1000 instances ≈ $309. The benchmark is therefore inexpensive to scale in API terms, an order of $300 takes it to 1000 instances, and the true constraint is the supply of candidate passages that contain structured, verifiable ambiguity. This favors *better passage pre-selection* (raising the 11.5% viability rate) over brute-force compute, and explains why the rare categories (e.g. filing_vintage), whose candidate passages almost never survive the constructor, are the expensive ones to grow.

H Construction and Validation Details

This appendix expands the data sources, curation pipeline, and validation summarized in §3.

Data sources. Figure 12 summarizes the resulting category and difficulty distributions. **English (EN, ∼80% of pool).** *DocFinQA*, 780 long-context passages extracted from 10-K and 10-Q filings, with verified candidate answers from annotated QA pairs. *EDGAR (XBRL-derived)*, 111 passages built from machine-readable XBRL facts for S&P 500 and Russell 1000 companies (FY2024–2025). **Chinese (ZH, ∼20% of pool).** *CNINFO/akshare*, 237 passages derived from A-share annual reports (SSE 50 + CSI 300) via structured financial-statement APIs, in three passage types: (1) metric definition, *kou-fei* vs. *gui-mu* (non-recurring-excluded profit vs. parent-attributable profit); (2) entity scope, *gui-mu* vs. *jing-lirun* (parent-attributable vs. fully consolidated including minority interest); (3) temporal scope, year-over-year revenue growth. ZH instances are expected to be harder for current models due to sparser training data on A-share filings and more complex corporate-structure nomenclature (e.g., A/H-share distinctions, *jituan* (group) vs. *zi-gongsi* (subsidiary) scope).

Three-role construction pipeline. Step 1: Constructor (GPT-5). A single GPT-5 call generates the full (Q, C, A) triple given a filing passage, verified candidate answer, and mandatory primary category. The output JSON includes `question`, `context`, `answer`, `default_answer`, `intended_evidence_span`, `default_evidence_span`, `intended_interpretation`, `default_interpretation`, and `axes_exercised`, with category-specific instructions injected per primary category. **Step 2: Pre-verifier sanity checks.** Fourteen automated code-level rules (Table 15) are applied without API calls, and any failure immediately rejects the passage. **Step 3: Adversarial verifier.** Ten verifier calls run in parallel (five GPT-5-mini and five GPT-5); each answers Q without C and states its assumed interpretation, and an instance is rejected if ≥ 2 of the 10 both produce A and align with the intended interpretation. This two-condition criterion prevents false rejections from lucky guesses.

Category diversity enforcement. We enforce target category shares through four mechanisms: (1) a priority function with a 3× penalty on over-quota categories; (2) hard exclusion at 2× quota; (3) a rarity-adjusted initial sort (passages sorted by $\sum_{\text{categories}}(\text{target\_share}/\text{pool\_frequency})$); and (4) a dynamic re-sort every 10 instances.

Human validation protocol. Two finance-literate annotators (one bilingual, for the Chinese split) independently review instances on an eight-question protocol: H1 *Is Q ambiguous without C?* H2 *Is the default interpretation plausible for a non-expert?* H3 *Does C uniquely identify the intended interpretation?* H4 *Does C avoid directly stating the answer?* H5 *Is A correct under the intended interpretation?* H6 *Is A_d correct under the default interpretation?* H7 *Is the primary category label correct?* H8 *What yes/no question would a financial analyst naturally ask to resolve the ambiguity?* (free text, used to compute human AC@1). Acceptance requires H1–H6 to hold after adjudication by the primary author. On a stratified 60-instance sample (by language and category), per-item raw agreement is 0.82–0.97. Because the labels are heavily skewed toward "yes" ($> 85\%$ per item), Cohen's κ is degenerate under the Feinstein–Cicchetti prevalence paradox, so we report Gwet's AC1 (0.85 average, range 0.78–0.97; 1.00 for H7). The single consensus-rejected instance (consensus rejection 1.7%) was a category over-reach, a revenue-recognition-timing pairing whose plausible default is the firm's total revenue rather than a disaggregation component. It is answerable only under context-oracle, where removing it shifts the ceiling by < 0.05pp (93–95% unchanged).

I Construction Quality-Control Rules

The pre-verifier applies fourteen automated code-level rules before any API call (Table 15), rejecting a candidate the moment any rule fails.

J Reproducibility

Models and decoding. Closed models are accessed via their official APIs (evaluated 2026): `gpt-5`, `gpt-4o`, `gpt-5-mini`. Reasoning models (gpt-5 family) use default temperature with a generous `max_completion_tokens` headroom (+4096 above the output budget) so internal reasoning does not truncate the answer; other models use temperature 0 for the agent and grader. **User simulator.** GPT-5 at temperature 1.0, constrained to {Yes, No, I don't know}

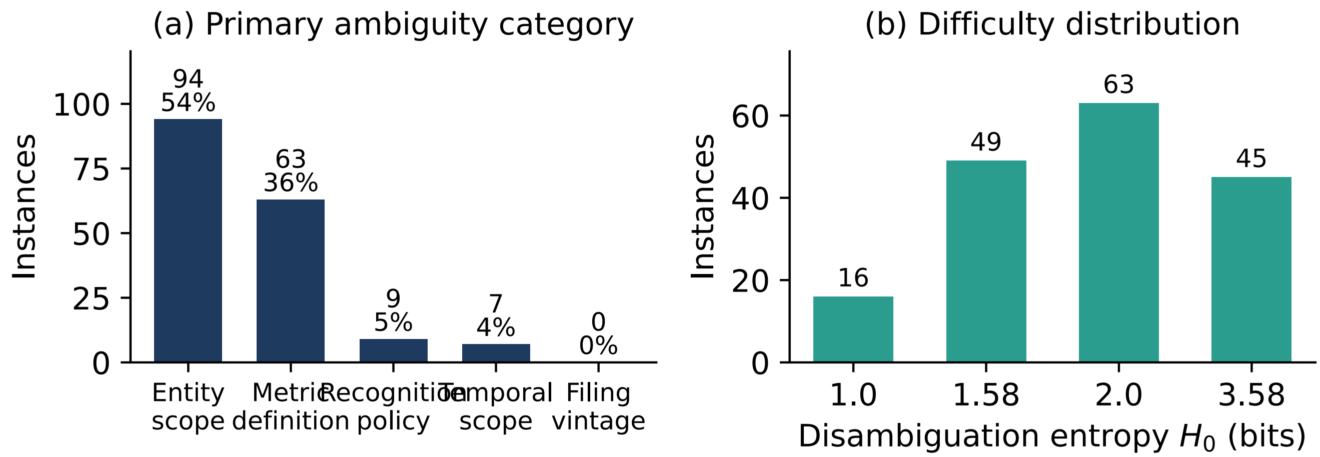

Figure 12: **FinInteract distributions.** *(a)* Primary ambiguity category: the corpus is entity/metric-dominant (54% / 36%), with the three subordinate categories sparse, mirroring how often each category is the *primary* ambiguity in public filings (Sec. 2). *(b)* Difficulty by disambiguation entropy $H_0 = \log_2 k$: instances span one to 3.58 bits, most at one or two bits with a substantial multi-category tail at 3.58 bits (45 instances).

Table 15: Pre-verifier sanity checks (14 rules).

Rule	Description
R1	Q must not be answerable yes/no
R2	Q must not contain disambiguating terms or in-line dates
R3	A taken verbatim from the source candidate value (not invented)
R4	Q must name the company
R5	Q must ask about a substantive financial metric
R6	Answer type consistent with question type
R7	Context C must not contain the answer value
R8	Intended and default interpretations must differ
R9	All category labels must be valid vocabulary items
R10	Answer must be non-trivial (not empty or "0")
R11	`default_answer` present and differs from `answer`
R12	Both evidence spans non-empty
R13	Evidence spans must be meaningfully distinct ($\leq$85% overlap)
R14	Intended and default *answers* differ beyond grader tolerance

and conditioned on the disambiguating context C. **Grader.** GPT-4o-mini at temperature 0, binary correctness with finance tolerance ($\pm 1\%$ numeric, entity/ticker equivalence, currency normalization; fiscal-year mismatch = wrong). **Protocol.** ReAct with `search`/`interact`/`answer` actions, max 10 rounds; the forced-interaction ablation requires $n{=}4$ asks before answering; the context-oracle ceiling supplies C plus both evidence spans (position-shuffled). **Statistics.** Accuracy CIs are 95% bootstrap over instances ($B{=}2000$); mode transitions use a two-sided paired bootstrap ($B{=}5000$). All evaluation, construction, and analysis scripts are released.

K Datasheet and Licensing

Provenance. English instances derive from public SEC EDGAR filings (10-K, FY2022–2025) and a small DocFinQA-sourced set; Chinese instances derive from public CSRC annual reports retrieved via CNINFO/akshare. All source documents are public regulatory filings; no proprietary, paywalled, or personal data is used. **Schema.** Each instance is a JSON record with fields: `instance_id`, `language`, `source`, `ticker`, `company`, `filing_type`, `filing_date`, `question`, `context` (C), `answer`, `default_answer`, `intended_evidence_span`, `default_evidence_span`, `intended_interpretation`, `default_interpretation` ({entity, period, metric, basis}), `axes`, `n_axes`, `h0`, and `qc` (per-rule pass flags + verifier blind-solve rate). **Intended use.** Evaluating whether interactive search agents recognize and resolve query ambiguity; not a training corpus and not financial advice. **License and release.** The benchmark, construction pipeline, and evaluation code are released for research use; redistribution of source filings follows the respective public-domain (EDGAR) and regulatory-disclosure (CSRC) terms. **Maintenance.** The frozen v1 release is versioned; corrections and the planned EN-split expansion will be issued as numbered releases.

Disclosure of Generative AI Use

Generative AI tools were used in this work in two distinct roles. First, as *components of the research artifact itself*: large language models act as the adversarial constructor, the user simulator, and the automated grader described in Section 3 and Section 4, and every such use is documented in the paper and the accompanying code with the specific model, version, and prompt. Second, generative AI tools were used to assist with *manuscript preparation*, limited to language editing, LaTeX formatting, and code refactoring. All research questions, experimental design, analyses, and scientific claims are the authors' own, and the authors have independently verified every result, table, figure, and citation reported here. The authors take full responsibility for the entire content of this paper, in accordance with the AAAI Code of Professional Ethics and Conduct.